

A Mean-Field Game of Mayoral Endorsements

Identification and Evidence from Brazilian Coalitional Presidentialism

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The problem

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It has gone structurally unestimated, because a discrete-choice game of incomplete information among Brazil's 5,570 mayors spans

$$(J+1)^I = 4^{5,570} > 10^{3,300} \text{ action profiles.}$$

Su–Judd, Aguirregabiria–Mira and Pesendorfer–Schmidt–Dengler all enumerate this space. Applications stay small.

What we change

Not the algorithm. The equilibrium concept.

Anonymity

A mayor's payoff depends on what others do only through the aggregate shares s , never through who in particular endorsed whom.

Equilibrium becomes a fixed point on the candidate simplex, of dimension J **whatever / may be**. The strategic content survives: each mayor still cares what the others do, through the share.

Existence by Brouwer. A sufficient condition for uniqueness holds in the region the data pick out. One estimation takes 0.3 ms.

Inverting the best response gives, for each type τ ,

$$\underbrace{\log \frac{\sigma(j^* | \tau)}{\sigma(0 | \tau)}}_{y_\tau} = X(\tau)' \beta + \alpha \cdot \frac{B_{j^*}}{s_{j^*}}.$$

A BLP-style inversion. Two-step OLS: K observations, $n_X + 1$ parameters.

Monte Carlo at Brazilian scale ($K = 30$, $n = 5,570$, $R = 500$): $\sqrt{n}$ consistency, near-nominal coverage under structural-delta standard errors.

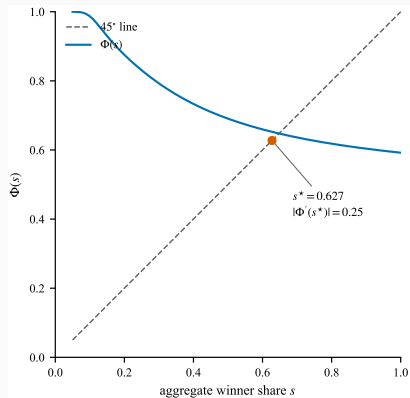
Dilma Rousseff's PT-led coalition, eight major parties.

Types: state $\times$ Power-Zucco bloc $\times$ within-state population tercile. $K = 222$ types, 5,441 mayors.

Preferred specification	
$\hat{\alpha} \cdot \hat{B}$	0.647
PSD s.e.	(0.051)
t	12.69

Precise. We now spend the rest of the talk on what it is precise **about**.

The equilibrium we estimate



Φ slopes down: a broader anticipated coalition dilutes B/s . Single crossing at $s^* = 0.627$, $|\Phi'| = 0.25 < 1$.

Now the part we would rather you heard from us

What that number is

Within a cycle the prize regressor is the **constant** column $1/\hat{s}_{j^*}$. The least-squares coefficient on a constant column obeys an identity:

$$\hat{\alpha} \cdot \hat{B} = \hat{s}_{j^*} \cdot b$$

where b is the intercept of the same regression run on $X(\tau)$ and a plain constant.

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So $\hat{\alpha} \cdot \hat{B}$ is the average type-level log-odds of alignment, net of covariates, rescaled by the share of mayors **classified** as aligned.

Who counts as aligned is a choice

Same cycle, same 5,441 mayors, same types, same covariates. Only the assignment rule changes.

	Definition	N aligned	$\hat{s}_{j^*}$	$\hat{\alpha} \cdot \hat{B}$
A	first-round coalition (headline)	3,283	0.627	+0.646
B	second-round coalition	3,295	0.667	+2.649
C	co-partisan only (PT)	602	0.202	−0.798

The estimate changes sign exactly where the aligned share falls below the outside option. All three are significant at $p < 0.001$.

Where the identity comes from

The endorsement measure is a party-coalition proxy. In 2014 it assigns all 5,657 mayors to a candidate; zero abstain.

So the estimation defines

$$\hat{\sigma}(0 \mid \tau) \equiv 1 - \hat{\sigma}(j^* \mid \tau),$$

the share assigned to a rival, not to no one.

The denominator is the complement of the numerator, so y_τ is the logit of the type's aligned share, and the identity follows. Everything on the previous two slides is a consequence of this one data fact.

A claim we are withdrawing

An earlier draft said the estimated bloc effects reproduce the Power–Zucco ideological ordering, *information the estimator never sees*. It sees it. Same 5,441 mayors, 222 types:

Regress the endorsement indicator on...	R^2
26 party dummies	1.0000
4 bloc dummies	0.2614

Not correlated with party. A **deterministic function of party** — as it must be, the proxy being built from party coalition lists. The bloc dummies are a coarsening of the outcome's own determinant, so any partition of parties reproduces an ordering mechanically.

And the ordering is not ideology

Share endorsing the winner, by bloc. Bloc labels are held fixed across cycles.

Bloc	2014 (Dilma)	2022 (Lula)
left_pt	0.988	0.327
left_psb	0.000	0.850
center	0.649	0.019
right	0.492	0.071

In 2014 left_psb is centre-left and sits **below** the right. In 2022 it sits at the top, above left_pt. Same partition, opposite order.

What the blocs order is coalition membership in that cycle. The paper's one externally checkable claim does not survive; we drop it rather than qualify it.

This also binds the scope conditions

We state four ex-ante criteria for where the model applies. Criterion (a) requires a winning coalition broad enough to support prize-divisible allocation.

Broad coalition $\Rightarrow \hat{s}_{j^*}$ large $\Rightarrow$ by the identity, $\hat{\alpha} \cdot \hat{B} > 0$.

The criteria are defensible as a statement of when the model's economics apply. They cannot *also* be an independent screen that the positive estimate then passes.

The negative estimates out of scope ($\hat{s}_{j^*} = 0.02$ in 2018, 0.11 in 2022) are the same arithmetic across cycles rather than a regime finding.

What does not depend on the model

Close-race DID, 2014 and 2022 pooled. Following Slough, Urpelainen and Yang (2017), alignment splits into *Coalition* (broad) and *CoPartisan* (narrow).

	Emendas (discretionary)	FPM (formula)
$\hat{\beta}_1$ Coalition	+0.007 (0.028)	−0.012 (0.009)
$\hat{\beta}_2$ CoPartisan	+0.106* (0.044)	+0.021 (0.016)
N muni-cycle	11,176	11,176

Premium on the discretionary channel, nothing on the formula-based one. The institution supplies its own placebo.

Robustness of the DID

- State-level clustering: $p = 0.039$
- Pre-trends, emendas and FPM: sharply null
- Inverse hyperbolic sine: +0.103; winsorised at 99th: +0.115
- 2018 wide-margin falsification: null
- Within-mayor switchers (306 mayors, mayor FE): +0.094, +0.108, +0.106 across specifications

The within-mayor rows lose power, not magnitude: only 23 CoPartisan-treated mayor-cycles.

New: how good is the proxy? (preliminary)

Mayors do not donate to presidential campaigns. But 458 of them donate to governors, senators and deputies, whose party is recorded. A donation is costly and individual.

Mayors with usable donations	446
Proxy agrees with donation-revealed alignment	316 (70.9%)
Donations going to the mayor's own party	57.7%

The error is not random. PT 93.2%, PCdoB 100%, PMDB 78.7% vs PTB 29.6%, PSC 41.7%, PSDB 58.1%: accurate for disciplined parties, poor for loose federations.

Stands The mean-field reduction and the tractability result. The DID premium and its placebo. The Monte Carlo.

Reframed $\hat{\alpha} \cdot \hat{B}$ is reported as an identity, and as a warning about what this class of estimator returns on a single cross-section.

Withdrawn The bloc-effect validation. The proxy is a deterministic function of party ($R^2 = 1.000$), so the estimator sees the ideology it appeared to recover.

Open Separating prize responsiveness from the composition of the alignment measure.

Two routes forward

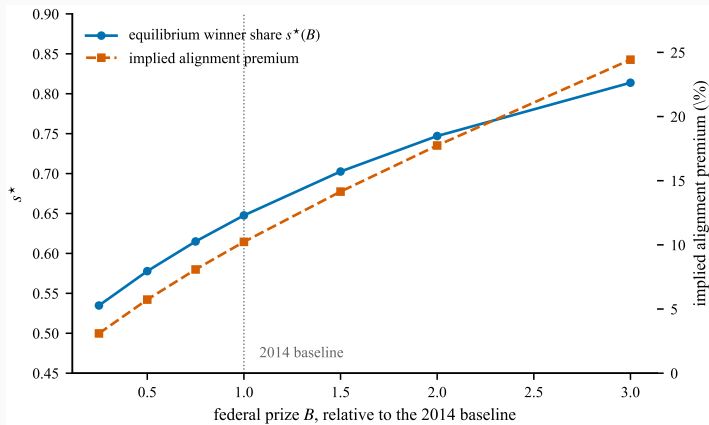
Both need variation in B/s that is not variation in a definition.

1. **Extend the TSE panel** to 2002, 2006 and 2010. Four *in-scope* cycles instead of one, with genuine variation in $\hat{s}_{j*}$ across them, and no need to reopen the scope decision.
2. **Multi-class with common β** , where the losing candidates' equations make $\hat{\alpha} \cdot \hat{B}$ a winner-versus-loser contrast rather than an intercept. This one needs **no new data**: the proxy already carries 8–12 candidates per cycle, and the binary collapse to endorses-winner throws them away. In 2014 the prize regressor would take the values 1.67, 4.04, 8.72, 54, ... instead of one number.

The alternative-specific β_j extension does **not** do this: with no constant in $X(\tau)$, $\hat{\alpha} \cdot \hat{B}$ is again the winner equation's intercept.

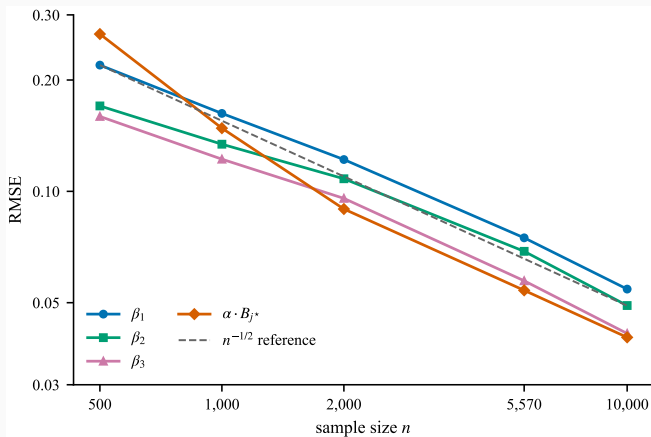
Thank you

Appendix: counterfactual prize sweep



Conditional on the identity above, the sweep shows the model's internal logic: a larger prize draws a broader coalition, which dilutes what each member receives, so the premium grows sublinearly.

Appendix: Monte Carlo convergence



Appendix: the four scope criteria

1. Coalitional presidentialism is the operating regime
2. Federal discretionary budget not bound by a fiscal rule (pre-EC 95/2016)
3. Patronage networks mature enough to forecast the prize
4. No extraordinary fiscal event (Lava Jato, COVID)

Of the Brazilian cycles observed, only 2014 satisfies all four. See slide on what this implies for the sign of $\hat{\alpha} \cdot \hat{B}$.